R1: Design Draft

EEL4907 CpE Design 1

Hardened RISC-V Exam Computer

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1 Introduction

1.1 Purpose / Need

The project aims to create a **plug-and-play secure exam computer** built around a RISC-V board. The motivation comes from the growing problem of AI-assisted cheating, especially in coding exams, and the limitations of current solutions such as Honorlock, which require intrusive monitoring of student laptops.

Instead of securing arbitrary student machines, we distribute **dedicated exam devices** during tests. Each device boots into a locked-down Linux environment and launches Safe Exam Browser (SEB) [1] so students take exams through Canvas. Because the hardware and software stack are controlled, tampering is much harder while avoiding the privacy concerns of remote proctoring tools. The system is needed by institutions and instructors who want exam integrity without surveilling students' personal devices.

1.2 Domain & Prior Work

The project sits at the intersection of secure boot, locked-down computing environments, and academic assessment. Prior work includes commercial proctoring (e.g., Honorlock, Respondus) and locked browsers (Safe Exam Browser). Our approach differs by controlling the entire device rather than layering software on untrusted student hardware.

Current status / prior work: This project was previously advanced by Nicholas DeSanctis. The RISC-V board already boots into a full Linux OS. Other students (Shane and Wilson) are integrating SEB with Canvas; our work will connect the RISC-V system to that pipeline.

1.3 Impact & Risk Assessment

The proposed hardened RISC-V exam computer addresses critical challenges in modern academic assessment, balancing security requirements with ethical and environmental considerations.

Social & Cultural: The primary social impact is the restoration of pedagogical integrity in an era where Large Language Models (LLMs) and generative AI can be deployed covertly to bypass traditional assessment. By hardening the testing environment, the project reinstates the incentive for students to master course material rather than relying on external automation. Culturally, this maintains the prestige of the University of Florida degree, ensuring that graduation signifies genuine competency and providing a level playing field for all students.

Economic: The current status quo for exam integrity relies on Software-as-a-Service (SaaS) models like Honorlock or Respondus. These solutions involve high operational expenditure, often reaching six-figure annual fees due to per-student, per-exam pricing structures. In contrast, this project represents a shift toward capital expenditure while the upfront hardware cost is higher, it eliminates recurring licensing fees. Additionally, maintaining a university-owned fleet creates technical student-employment opportunities, providing hands-on experience in open-source maintenance/development and hardware maintenance.

Ethical: Modern proctoring software often requires intrusive access to student-owned hardware and personal environments, raising significant privacy concerns. This project addresses these ethical dilemmas through a design-for-privacy approach, shifting the security burden from the student's personal device to university-owned hardware. This ensures that no invasive monitoring

software—which can often act as a rootkit—is ever installed on a student's personal property.

Environmental & Global: Utilizing the RISC-V architecture promotes an open-standard Instruction Set Architecture (ISA), reducing global reliance on proprietary silicon. From an environmental standpoint, these boards are high-efficiency, low-power devices (typically <15W), significantly reducing the aggregate carbon footprint compared to a classroom of high-performance consumer laptops. The use of 3D-printed enclosures further allows for localized manufacturing and the use of recyclable materials like PETG or PLA.

Risk Assessment & Limitations: The primary technical risk is the "arms race" of hardware exploits. Since students have physical access to the device, sophisticated actors could attempt to interface with the hardware via GPIO pins or physical keyloggers. Furthermore, as a custom solution, there is a risk of hardware failure during an active exam. To mitigate these risks, the design incorporates a secure boot chain to detect system tampering and a deployment strategy that replaces faulty units immediately without loss of student exam progress.

2 Statement of Work

Primary Deliverable: The development of a RISC-V native, SEB-compatible kiosk environment. Because official SEB support is restricted to Windows, macOS, and iOS, this project will implement a **Linux SEB-Wrapper**. This wrapper will utilize a Wayland-based Kiosk compositor to replicate the Kiosk application lockdown features and a custom browser instance to handle the Browser Exam Key (BEK) and Configuration Security Key (CSK) handshakes required by Canvas.

Performance Specifications:

- **SoC:** SpacemiT K1 (8-core RISC-V) with hardware-accelerated AES/SHA for configuration decryption.
- **Lockdown Efficacy:** System must disable all TTY switching (Ctrl+Alt+Fx), terminate non-whitelisted processes, and prevent escape via standard window management shortcuts.
- **Integrity:** Use RISC-V PMP (Physical Memory Protection) to isolate the SEB configuration keys from user-space memory.

ID	Task Description	Owner	Target Date
SW-01	Port SEB configuration parsing logic (XML/PList) to C++/Python for Linux.	Name	April ??
SW-02	Implement the SEB-to-LMS Handshake (BEK/CSK hashing) to satisfy Canvas security.	Name	April ??
UI-01	Configure Cage (Wayland compositor) for single-application kiosk mode.	Name	April ??
SEC-01	Implement Secure Boot and read-only eMMC partitioning on the BPI-F3.	Name	April ??

Table 1: Project Milestones for SEB-Linux Implementation

3 Deliverable Artifacts

3.1 Hardware

Description: A simple physical exam device for easy deployment (e.g., 3D-printed enclosure mounting the RISC-V board to the VESA mount on the back of a monitor, creating a compact "all-in-one" exam computer with only power, keyboard, and mouse).

Accessibility, Usability, & Maintenance: The hardware utilizes a standard VESA mount to ensure compatibility with existing university laboratory infrastructure. For maintenance, the enclosure is designed for tool-less or single-fastener access to allow for rapid replacement of the RISC-V board or storage media in the event of hardware failure. Usability is prioritized through a simplified I/O layout, reducing the potential for student error if connecting peripherals.

3.2 Software

Description: Hardened Linux image, secure boot components, and integration with Safe Exam Browser and Canvas.

Dissemination Plan: All produced software will be open sourced, barring components that must remain confidential to maintain the hardening integrity of the system. By design, the software will be packaged and integrated with the OS of the device and will require no additional building, installation, or configuration by the end-user.

Accessibility, Usability, & Maintenance: The software environment is designed to be self-healing; since the eMMC is mounted as a read-only filesystem, any accidental or malicious changes are wiped upon a power cycle. For accessibility, the kiosk browser will support standard web accessibility features, including screen reader compatibility via the ARIA standard and keyboard-only navigation, ensuring compliance with ADA requirements for testing environments.

3.3 Documentation

Description: Manuals, specifications, and reports (including this design draft and subsequent deliverables).

Dissemination Plan: All produced documentation will be open sourced alongside the project software. Additionally, documentation will be provided for use and procedures at each level of authority. That is, documentation will be provided for those who intend to build the project, professors who aim to use the device, and students requiring instructions on proper operation.

Accessibility, Usability, & Maintenance: All documentation will be provided in searchable, screen-reader-friendly PDF and Markdown formats to ensure accessibility for all users. Maintenance will be managed via a hosted documentation platform (e.g., ReadTheDocs or a GitHub Wiki), allowing for real-time updates as the RISC-V ecosystem or Canvas API evolves. Versioned manuals will ensure that documentation remains synchronized with the specific hardware revisions deployed in the field.

4 Mockups

4.1 Interfaces

TODO

4.2 Systems

TODO

4.3 Networking

TODO

4.4 Storyboards

The following describes the intended user workflow from device power-on to exam completion.

1. **Device Power-On:** The proctor distributes the exam devices and powers them on. No student interaction is required at this stage.
2. **Secure Boot:** The system verifies the integrity of the OS image via cryptographic hashes. If the image has been tampered with, the device halts and alerts the proctor.
3. **Hardened Linux Boot:** The device boots into the locked-down Linux environment. All unnecessary services, TTY switching, and window management shortcuts are disabled.
4. **SEB Launch:** Safe Exam Browser launches automatically in kiosk mode, filling the entire screen. The student cannot exit or access any other application.
5. **Canvas Login:** The student is presented with the Canvas login page inside SEB. They log in using their university credentials.
6. **Exam Session:** The student completes their exam through Canvas. The Browser Exam Key (BEK) and Configuration Security Key (CSK) handshakes ensure the session is valid and untampered.
7. **Session Wipe:** Upon exam completion or device power-off, the read-only filesystem ensures any temporary session data is wiped on the next power cycle.

4.5 Draft Schematics

TODO

5 Stretch Goals

(1) System for distributing configuration to exam devices (exam settings, network config). (2) Hardware-bound cryptographic keys per device (e.g., private key to decrypt/verify config), so the exam environment only runs on approved hardware. Additional hardening (kernel hardening, capability restriction, stronger secure boot) may be needed as the project progresses.

References

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[4] RISC-V International, "The RISC-V Instruction Set Manual, Volume II: Privileged Architecture," Document Version 20211203, Dec. 2021. [Online].
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